

Similarity-Aware Techniques for Deduplication in Large-Scale Remote Sensing Data

1st Yuchun Wang

School of Computer Science and Technology
Changchun University of Science and Technology
Changchun, China
3455716550@qq.com

2nd Guoying Zhang

School of Computer Science and Technology
Changchun University of Science and Technology
Changchun, China
zhanggy1103@126.com

3rd Jingqi Wang

School of Computer Science and Technology
Changchun University of Science and Technology
Changchun, China
wjingqi666@126.com

4th Hui Qi

School of Computer Science and Technology
Changchun University of Science and Technology
Changchun, China
qh_odd@126.com

5th Chunbo Wang*

School of Computer Science and Technology
Changchun University of Science and Technology
Changchun, China
wangcb1213@126.com

Abstract—With the continuous advancement of satellite remote sensing technology, the issue of data redundancy in remote sensing images caused by multi-satellite, multi-angle, and multi-temporal observations has become increasingly prominent. This redundancy not only reduces storage efficiency but also affects the performance of subsequent data processing. To address this problem, this paper proposes a Transformer-based joint mechanism for remote sensing image classification and similarity-based deduplication. A lightweight Vision Transformer model is designed based on embedded feature representation learning, which can effectively capture the global contextual and spatial structural information of images. This design significantly improves computational efficiency while maintaining high classification accuracy. Furthermore, the standardized cosine similarity is adopted to measure the feature similarity between images, and redundant data are effectively eliminated by applying an adaptive threshold. Experimental results demonstrate that the proposed method outperforms existing approaches in terms of feature extraction efficiency, classification accuracy, and deduplication performance. Therefore, it can effectively alleviate the storage and computational burdens caused by redundant remote sensing data and shows great potential for practical applications, particularly in fields such as urban planning and environmental monitoring.

Index Terms—Similarity Computation, Remote Sensing Data, Image Deduplication, Lightweight Transformer.

I. INTRODUCTION

With the rapid development of satellite remote sensing technology, multi-satellite cooperative Earth observation has been widely applied in fields such as environmental monitoring, urban planning, agricultural management, and

disaster assessment [1]. However, as illustrated in Figure 1, different satellites may capture images of the same ground area from different viewing angles within a similar time window, leading to a high degree of spatial redundancy among the acquired remote sensing images. In addition, as shown in Figure 2, when multiple satellites conduct multi-temporal observations of the same region at different time intervals, significant temporal redundancy arises. The data redundancy jointly induced by multi-satellite, multi-angle, and multi-temporal observations has become increasingly pronounced, severely constraining both the storage efficiency and processing performance of remote sensing data [2]. Such redundancy not only leads to a substantial increase in storage and transmission costs but may also obscure critical features during data analysis, thereby affecting both the training efficiency and accuracy of learning models [3]. In current data management frameworks, remote sensing images are typically transmitted and centrally stored in ground-based or cloud data centers. In the absence of effective similarity detection mechanisms, the massive accumulation of redundant images results in wasted storage resources and imposes a heavier burden on data transmission and retrieval systems. Moreover, the continuous accumulation of duplicate data contradicts the prevailing objectives of green computing and resource efficiency. Therefore, developing similarity-based deduplication techniques for remote sensing imagery is of great theoretical significance and practical value, as it can substantially improve overall system performance and data management efficiency.

In image deduplication techniques for blurred or near

Corresponding author: Chunbo Wang (wangcb1213@126.com)

duplicate images, those that appear almost identical to the human visual system but differ at the byte level are likewise regarded as redundant data and included in the deduplication scope. By retaining only a single representative copy of similar images in the cloud, additional storage and communication overhead can be effectively reduced [4]. Following a similar principle, recent research has increasingly focused on leveraging deep learning models to capture the semantic similarity between images. Yu et al. compared several Transformer-based architectures—such as SETRnet, SwinUnet, and TransUnet—in remote sensing image segmentation tasks. Their results demonstrated that Transformer models significantly outperform traditional convolutional neural networks (CNNs) in representing large-scale semantic features [5]. Furthermore, Aleissae et al. highlighted in their review that the Vision Transformer (ViT) and its variants can effectively capture long-range dependencies and global contextual information within images, exhibiting outstanding modeling capability when dealing with ultra-high-resolution imagery and heterogeneous land-cover scenes [6]. In summary, semantic-level similarity modeling based on deep learning provides a novel technical pathway for intelligent deduplication of remote sensing imagery, marking a paradigm shift from “perceptual-level deduplication” toward “semantic-level understanding.”



Fig. 1: Simultaneous multi-satellite observations from different angles.



Fig. 2: Multi-Satellite Observations of the Same Location at Different Times.

In the field of image processing, the Transformer architecture has been extensively validated for its remarkable ability to model long-range dependencies and semantic similarity among images, demonstrating superior performance in tasks such as image classification, retrieval, and compression [7]. Particularly in remote sensing image processing, recent studies have explored the use of models such as the ViT to extract scene-level semantic features, thereby enhancing the understanding and expressive capability of remote sensing data [8]. However, in the specific context of remote sensing image similarity-based deduplication, how to effectively integrate Transformer-based models to achieve fast semantic classification and efficient redundancy elimination for large-scale image datasets remains an open and challenging research problem that warrants further investigation.

Although remote sensing images typically contain abundant spatiotemporal reference information, which can assist in identifying similar images to some extent, achieving efficient and accurate deduplication in practice remains challenging due to complex factors such as cloud cover, illumination variations, and land-cover heterogeneity. Moreover, similar images from different regions may need to be retained for specific research purposes, further increasing the complexity of deduplication decisions. Therefore, a key issue in improving deduplication performance lies in how to adaptively determine similarity thresholds while maintaining high classification accuracy. To address these challenges, this paper proposes a Transformer-based integrated mechanism for classification and similarity-based deduplication. A lightweight Vision Transformer model is introduced to perform deep feature extraction and redundancy identification on remote sensing images, enabling efficient semantic-level deduplication. By retaining only representative or unique information segments, the proposed method significantly reduces storage and computational overhead, while ensuring effective data reduction from a semantic understanding perspective. The main contributions of this paper can be summarized as follows:

- This paper proposes a Transformer-based joint mechanism for classification and similarity-based deduplication, following a “classification-then-deduplication” workflow. By integrating feature extraction, classification, and deduplication modules, the method establishes a complete pipeline from feature learning to redundancy elimination. The proposed approach significantly enhances feature discriminability and outperforms traditional methods in terms of both classification accuracy and deduplication efficiency, providing an effective solution for efficient management of remote sensing data.
- A lightweight Vision Transformer model based on embedded feature representation learning is designed to effectively capture the global contextual and spatial structural information of images. Through a

comprehensive and in-depth evaluation on remote sensing image classification tasks, the proposed method achieves high classification accuracy while significantly improving model efficiency.

- Furthermore, a threshold-based deduplication mechanism employing standardized cosine similarity is integrated to achieve precise redundancy identification, with the threshold value determined through extensive experimentation. This mechanism significantly enhances both deduplication efficiency and storage utilization efficiency.

The remainder of this paper is organized as follows: Section II reviews related work by other researchers on secure deduplication techniques. Section III provides a detailed introduction to the proposed scheme framework and the entire workflow. Section IV analyzes the experimental results. Section V summarizes the research work of this paper.

II. RELATED WORK

With the surge in satellite remote sensing image data, the issue of data redundancy has become increasingly severe, leading to increased storage and computational costs while affecting the efficiency and accuracy of data analysis. Remote sensing image deduplication techniques, by extracting visual or semantic features for efficient matching, provide an effective approach to optimizing data utilization. Early content-based image retrieval (CBIR) methods, such as handcrafted features like HOG and LBP, achieved partial image matching but faced limitations in expressive power when handling the multi-scale and complex structures of remote sensing images. Deep learning methods, particularly CNN-based hash learning and Vision Transformer (ViT)-based semantic embedding, have significantly improved retrieval performance by generating compact feature vectors that enhance matching accuracy. However, these methods still face challenges in addressing image blurriness and uncertainty. To tackle this, Lu et al. [9] proposed the DFHN method, which combines fuzzy logic and deep neural networks to address the aforementioned issues. Additionally, some studies have explored integrating traditional transforms with deep learning. For example, Singh et al. [10] proposed using Haar wavelet transforms to extract edge detail features, combined with CNN for feature learning, successfully applied to near-duplicate detection in multimedia images, demonstrating the potential of designing feature extraction strategies for specific image differences. Multimodal retrieval methods, by integrating images with metadata (e.g., geographic coordinates, timestamps) [11], [12], enhance the processing capability for complex scenes, offering valuable insights for deduplication tasks. However, these methods typically rely on high bandwidth and abundant computational resources, making it challenging to meet the real-time requirements of low-orbit satellite networks. Vision Transformers, with their global modeling capabilities, have shown superior performance in remote sensing image processing, particularly in scene classification [13] and change detection [14], improving performance through

temporal dependency modeling. Multimodal Transformer models further integrate images and metadata, enhancing the analysis of complex scenes [15]. Wang et al. [16] proposed a novel deep hashing method based on a cross-scale Transformer structure to extract multi-scale image features, aiming to improve image retrieval performance. These studies indicate that Transformers' advantages in semantic feature extraction and multimodal processing can effectively support deduplication tasks, particularly for handling visually similar but byte-stream-different remote sensing images. However, existing research primarily focuses on classification and segmentation, and their application to bandwidth-constrained deduplication scenarios remains underexplored. In recent years, Gardère et al. [17] proposed a method for automatically constructing large-scale fashion synthetic image retrieval (CIR) datasets and introduced a novel CIR model. By pretraining on this dataset, they significantly improved performance in fashion CIR tasks, particularly in fine-grained text-modified retrieval. This suggests that pretrained deep learning models and dataset construction strategies can effectively enhance image retrieval performance, providing a new technical pathway for remote sensing image deduplication tasks.

Image deduplication, by identifying visually similar images to reduce redundant data, is a critical method for optimizing the storage and processing of remote sensing images. Traditional approaches primarily rely on low-level visual feature similarity metrics, such as perceptual hashing (pHash) [18] and Scale-Invariant Feature Transform (SIFT) [19], as well as deep learning methods like Siamese networks [20] and triplet loss [21]. In recent years, multimodal data exploration methods have also been introduced to the field of image deduplication. Thus, multimodal data exploration methods provide new ideas and technical approaches for remote sensing image deduplication. However, the local receptive field of convolutional neural networks (CNNs) limits their ability to capture long-range dependencies. Cloud-based deduplication research, through perceptual hashing systems and deep learning frameworks, detects near-duplicate images, optimizing storage efficiency and improving deduplication performance. Multimodal deduplication methods enhance accuracy by integrating images with metadata (e.g., geographic coordinates, timestamps), particularly through hashing methods based on fuzzy logic and deep learning. Additionally, Du et al. [22] proposed an image deduplication algorithm based on self-supervised deep hashing, utilizing the SwinFpnHashCLR model to generate multi-scale feature hash codes and combining Hamming distance with the Otsu algorithm for efficient deduplication to address image redundancy. Luong et al. [23] introduced a patch-based deduplication strategy, PXDedup, which effectively identifies and eliminates visual redundancy. However, software implementations on standard CPUs suffer from low throughput due to processing limitations. Moreover, cloud-based deduplication relies on high bandwidth and abundant computational resources,

making it challenging to meet the real-time and bandwidth constraints of low-orbit satellite networks. For forensic images in mobile crowdsensing vehicular networks, Li et al. [24] proposed a secure near-duplicate deduplication method. By integrating cloud-fog collaboration and encryption techniques, this method effectively detects and removes near-duplicate images while preserving privacy, offering a novel solution for achieving efficient and secure image deduplication in resource-constrained environments.

Remote sensing image deduplication faces significant challenges arising from high resolution, multi-scale characteristics, and complex scenarios such as cloud cover, illumination variation, and land-cover heterogeneity. Traditional methods, such as perceptual hashing (pHash) and SIFT, exhibit low accuracy under noise interference, while CNN-based approaches are limited by their local receptive fields, making it difficult to capture long-range dependencies. Although the ViT enhances feature extraction capability, its computational complexity and resource demands hinder its applicability in low Earth orbit (LEO) satellite networks, where bandwidth and real-time constraints are critical. Multimodal deduplication methods, which integrate image data with metadata, can improve precision but still face challenges in adapting to heterogeneous remote sensing scenarios. Moreover, existing studies lack systematic solutions tailored for resource-constrained environments, making it difficult to balance storage efficiency with the goals of green computing. To address these issues, this paper proposes a ViT-based joint mechanism for remote sensing image classification and similarity-based deduplication. By combining a lightweight Transformer architecture with a cosine similarity-based threshold deduplication mechanism, the proposed method enables efficient and robust redundancy identification and elimination. The scheme significantly reduces storage and transmission costs in LEO satellite networks, providing an innovative solution for remote sensing data management and energy-efficient (green) computing.

III. METHODOLOGY DESIGN

This paper proposes a remote sensing image similarity-based deduplication framework based on a lightweight Vision Transformer, as illustrated in Figure 3. In the proposed approach, the positional encoding, CLS token, and deep stacked layers are removed from the standard Vision Transformer architecture. The framework first employs a Patch Embedding module to divide the input image into 1010 non-overlapping patches (a total of 484) and projects them into a 768-dimensional embedding space via convolution. Subsequently, a single-layer Transformer encoder (comprising 12-head self-attention and a feed-forward network) processes the patch sequence. To achieve lightweight design, positional encoding and the CLS token are removed, and GAP is applied to aggregate all patch features into a single image-level vector. Finally, this vector is used for classification (via a two-layer MLP) and similarity computation (using normalized cosine similarity) to eliminate redundancy based on a threshold.

This design effectively preserves semantic representation capability while compressing the model parameters to 1.8M and significantly improving inference speed.

A. Feature Extraction

The core of the feature extraction module is a lightweight variant of the Vision Transformer. First, the input image is divided into 10×10 patches and projected into a 768-dimensional embedding space. Compared with the conventional 16×16 patch size in standard ViT, this smaller patch configuration effectively preserves fine-grained local details. Subsequently, the patch sequence is processed by a single-layer Transformer encoder to capture global dependencies. To reduce computational overhead, positional encoding is omitted since in remote sensing imagery, global semantics (e.g., land-cover types) dominate classification performance, while positional information is often redundant. Meanwhile, GAP is adopted to replace the CLS token, directly averaging all patch embeddings, which enhances robustness and avoids sequence bottlenecks. Finally, an optional fully connected layer ($768 \rightarrow 768$) can be added to strengthen nonlinear representation capability. The module outputs a compact 768 dimensional feature vector for downstream tasks.

The multi-head attention mechanism enables the model to learn features from multiple representation subspaces in parallel, allowing it to interpret image content from diverse perspectives and thereby enhancing the richness and stability of semantic features. This module ultimately produces high-quality feature embeddings suitable for downstream tasks such as image retrieval, redundancy elimination, and scene understanding, demonstrating strong generality and expressive capability in remote sensing image analysis.

B. Classification

The classification module performs remote sensing scene classification based on the extracted feature vectors using a two-layer MLP ($768 \rightarrow 384 \rightarrow 19$). This design is lightweight and efficient, with the ReLU activation function enhancing the model's nonlinear representation capability. Experimental results demonstrate a classification accuracy of 98.36%. Benefiting from the global attention mechanism of the Transformer, the model can effectively distinguish visually similar scenes even without positional encoding.

It is worth emphasizing that the design of the classification module serves not only for semantic label prediction, but more importantly, to guide the Transformer in learning a well-structured embedding space. In this space, semantically similar images tend to cluster together, while dissimilar samples remain clearly separated. This property is particularly critical for identifying and eliminating redundant information in remote sensing imagery, while also enhancing the system's generalization and robustness in downstream tasks such as image retrieval and scene understanding.

C. Similarity Computation and Deduplication

In the similarity computation module, the feature vectors extracted by the Transformer are normalized to unit norm

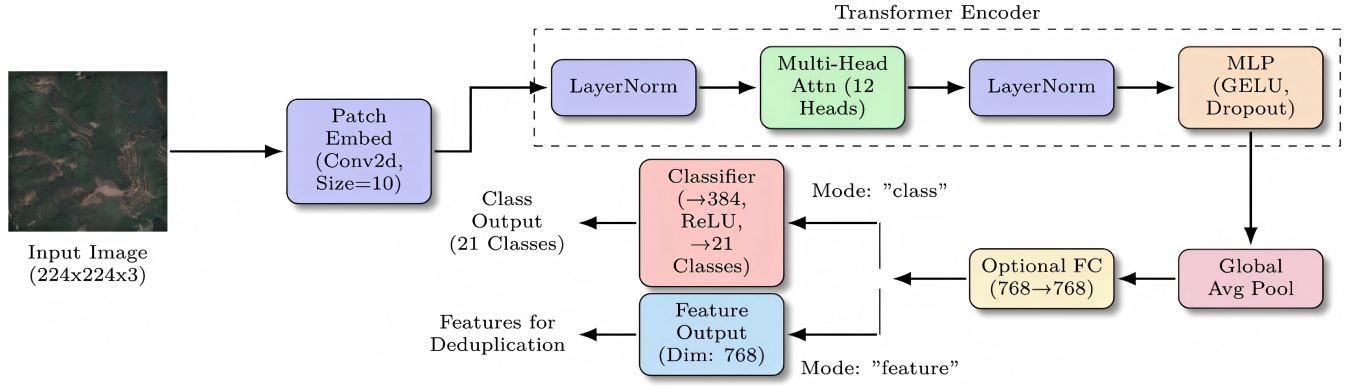


Fig. 3: Structural diagram of the lightweight Vision Transformer model for remote sensing image feature extraction.

to eliminate scale differences. Cosine similarity is used to measure the semantic similarity between image pairs:

Cosine Similarity for Similarity Determination:

$$\text{cosine}_{\text{similarity}}(x, y) = \frac{x \cdot y}{\|x\| \cdot \|y\|}$$

where f_i and f_j represent the feature vectors of images i and j , derived from the feature patterns output by the Transformer. Most features of remote sensing images are non-negative, with output values ranging between $[0, 1]$, where values closer to 1 indicate higher similarity.

A threshold τ is set, and if the condition is satisfied:

$$\text{cosine}_{\text{similarity}}(f_i, f_j) > \tau$$

then images i and j are considered duplicates and can be deduplicated.

After the image classification stage, the model generates a fixed-dimensional semantic feature vector for each image, serving as its compact semantic representation. To further eliminate redundant samples from the classification results, this paper performs pairwise similarity computation within each category based on the feature vectors $f_i \in R^D$ extracted by the Transformer. Specifically, the cosine similarity between any two images is computed, and a threshold of $\tau = 0.92$ is determined through extensive experiments. When the similarity exceeds this threshold, the two images are identified as duplicates or highly redundant samples. The selection of this threshold is based on the following considerations: (1) Empirical setting: It is a commonly adopted empirical value in image deduplication tasks, offering a good trade-off between recall and precision. (2) High similarity requirement: A cosine similarity of 0.95 indicates that the two image feature vectors are extremely close, with only a very small angular deviation. (3) Reduction of false positives: A relatively high threshold helps prevent dissimilar images from being mistakenly identified as duplicates. Since cosine similarity is inherently robust to variations in brightness and scale, it can reliably capture the semantic structural consistency between images, thereby significantly improving

the redundancy detection accuracy during the deduplication process.

D. Model Analysis

1) *Model Evaluation:* In remote sensing image classification tasks, Accuracy reflects the overall prediction correctness of the model, while Macro-Precision, Macro-Recall, and Macro-F1 are designed for class-imbalanced scenarios, emphasizing the model's ability to learn from and balance small-sample classes. High classification performance ensures compact intra-class features and well-separated inter-class representations in the embedding space, thereby providing strong discriminative support for subsequent cosine similarity-based deduplication.

Accuracy:

$$\text{Accuracy} = \frac{\sum_c TP_c}{\sum_c (TP_c + FP_c + FN_c + TN_c)}$$

Precision:

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}$$

Recall:

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c}$$

F1 Score:

$$F1_c = 2 \cdot \frac{\text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}$$

where TP_c denotes the true positives of class c (predicted as c and actually belonging to c), FP_c represents the false positives of class c (predicted as c but actually not belonging to c), FN_c denotes the false negatives of class c (actually belonging to c but predicted as another class), and TN_c refers to the true negatives of class c (predicted as not c and actually not belonging to c). C denotes the total number of classes.

Class Center Distance Calculation: For each class, 30 samples are selected to compute the class center feature:

$$\mu_c = \frac{1}{N_c} \sum_{i=1}^{N_c} f_i$$

Inter-Class Euclidean Distance:

$$d_{ij} = \sqrt{\sum_{k=1}^D (\mu_{i,k} - \mu_{j,k})^2}$$

The distribution of class centers can reflect the presence of potential “fuzzy zones” in the embedding space, aiding in the identification of category boundary ambiguity. This helps explain why certain highly similar images, despite high classification accuracy, may still be misjudged by the deduplication algorithm as belonging to different categories. Such misjudgments typically arise when image features fall in boundary regions near class centers. Furthermore, a visualized threshold heatmap can be constructed based on the distance relationships between class centers, enabling dynamic adjustment of deduplication strategies for different categories to enhance the discriminative capability and adaptability of the deduplication system.

2) t-SNE Visualization Analysis:

$$L_{t-SNE} = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}}$$

p_{ij} : similarity between samples i and j in the original high-dimensional space (Gaussian distribution).

q_{ij} : similarity between samples i and j in the low-dimensional (2D) space (t-distribution).

- 1) t-SNE aims to preserve the neighborhood relationships in the high-dimensional space; therefore, if the original feature space is well-classified, distinct clusters should form in the t-SNE visualization.
- 2) Tight clustering $\rightarrow$ strong intra-class feature consistency; clear separation between clusters $\rightarrow$ good inter-class distinction.
- 3) If “similar images” cluster together while “dissimilar images” are distributed across different classes, it indicates that the features learned by the model are well-suited for semantic deduplication tasks.

t-SNE is a nonlinear dimensionality reduction method used to visualize the geometric structure and neighborhood relationships of feature vectors in the embedding space, preserving local structure to reflect the relative distribution of data points. In contrast, cosine similarity quantifies the proximity of feature vectors in high-dimensional space by comparing their directions. In t-SNE visualization, if the feature points of two images are close in the reduced-dimensional space, it typically indicates a high cosine similarity in the original high-dimensional space, reflecting similar feature directions. Thus, t-SNE serves as a visualization validation tool for the cosine similarity-based deduplication strategy, providing geometric intuition and discriminative evidence for image duplicate detection by intuitively displaying the clustering structure in the embedding space.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

To comprehensively validate the effectiveness of the proposed lightweight Vision Transformer in remote sensing image classification and similarity-based deduplication tasks, experiments are conducted on two publicly available benchmark datasets: WHU-RS19 [25] and the UCMerced Land Use Dataset (hereinafter referred to as UCMerced) [26].

The WHU-RS19 dataset comprises 19 land-cover categories (such as airport, bridge, and forest), containing a total of 1,005 high-resolution remote sensing images with a spatial size of 600×600 pixels, approximately 50 images per category. This dataset features a balanced class distribution and well-defined scene semantics, making it suitable for evaluating the model’s overall performance in terms of global classification accuracy.

The UCMerced dataset consists of 21 land-use categories (such as agriculture, airplane, and building), containing a total of 2,100 aerial images with a spatial resolution of 256×256 pixels, with 100 images per category. This dataset exhibits class imbalance and significant variations in illumination, scale, rotation, and viewpoint, making it more suitable for evaluating the model’s robustness and generalization capability under complex visual conditions. We adopt Macro-Precision, Macro-Recall, and Macro-F1 as evaluation metrics to emphasize the model’s learning performance on minority classes.

In the experiments, the WHU-RS19 dataset is employed to evaluate the backbone classification and deduplication performance, reporting the overall Accuracy and deduplication F1 score. The UCMerced dataset is used as a complementary validation set, focusing on analyzing the model’s stability under class imbalance and visual perturbations, with results reported in terms of Macro-averaged metrics.

A. Feature Extraction Speed Analysis

To validate the computational efficiency of the proposed lightweight Vision Transformer in remote sensing feature extraction, this section conducts experiments on the WHU-RS19 test set using 100 remote sensing images with a resolution of 224×224 pixels. We compare three methods: the customized lightweight Transformer, the conventional Vision Transformer (ViT-B/16), and the traditional Local Binary Pattern (LBP) feature extraction. All methods output 768-dimensional feature vectors to ensure comparable semantic representation capability and fair comparison. The experiments are conducted on a single NVIDIA GPU with CUDA support, using a batch size of 8.

1) *Experimental Setup*: **Lightweight Transformer**: A customized single-layer Transformer encoder is adopted with a 10×10 patch division and position-free encoding. The [CLS] token and GAP are fused, resulting in approximately 1.8M parameters and 768-dimensional output features.

Conventional ViT-B/16: The pre-trained model torchvision.models.vit_b_16 is used, where the classification head is removed, and the [CLS] token output (dimension

768) is taken as the feature representation. The parameter size is about 86M.

LBP Features: Based on the skimage library, the uniform Local Binary Pattern (LBP) is implemented with a radius of 3 and 24 neighboring pixels. The obtained histograms are normalized using PCA or zero-padding to a uniform 768-dimensional feature vector.

2) *Speed Comparison Results:* For the three methods, the total feature extraction time on 100 images and the average time per image are shown in Table IV.

TABLE I: Feature Extraction Speed Comparison (100 Images, 768-Dimension)

Method	Total Time (s)	Time per Image (ms)	Relative to Custom Model
Lightweight Transformer	0.458	4.58	1.00×
Conventional ViT-B/16	0.933	9.33	2.04×
LBP	5.892	58.92	12.87×

The custom model achieves the fastest speed: thanks to its single-layer attention mechanism and the minimalist design without positional encoding and CLS token, our model takes only 0.458 s for 100 images, or 4.58 ms per image. It is 2.04 times faster than the conventional ViT and 12.87 times faster than LBP, demonstrating a significant advantage in computational efficiency.

The efficiency bottleneck of conventional ViT is evident: although ViT-B/16 possesses stronger representational capacity, its 12-layer deep stacking and self-attention computation over a sequence length of 197 (196 patches + 1 CLS token) result in significantly higher inference overhead, taking approximately twice the time per image compared to our model.

LBP incurs the highest computational overhead: although conventional LBP requires no training, computing local patterns pixel by pixel and constructing histograms on the CPU is inherently serial, resulting in very low efficiency on high-resolution images. The per-image processing time reaches up to 58.92 ms, making it unsuitable for real-time or onboard processing scenarios.

Conclusion: The experimental results demonstrate that the proposed lightweight Vision Transformer achieves millisecond-level feature extraction (4.58 ms per image) while maintaining high-dimensional 768-D expressive features, significantly outperforming the conventional ViT and LBP methods.

B. Classification Performance Analysis

The classification task provides a solid foundation for the subsequent deduplication process by generating reliable category labels. In this work, the performance of the lightweight Vision Transformer model was systematically evaluated based on classification accuracy and loss. To comprehensively verify the model's classification capability across different datasets, as well as to assess the impact

of lightweight design choices such as removing positional encoding on classification performance, we conducted comparative experiments with several mainstream remote sensing image classification methods. The classification accuracy results on the WHU-RS19 dataset are presented in Table II.

TABLE II: Comparison of Classification Performance of Different Models on the WHU-RS19 Dataset

Methods	Accuracy (%)
B-CNN [27]	82.83
ResNeXt-50 [28]	83.27
TAE-Net [29]	88.04
Ours	98.36

The results indicate that removing the positional encoding can still maintain high classification accuracy. Therefore, positional encoding is not essential for remote sensing image classification tasks.

Although the standard Vision Transformer relies on positional encoding to model the spatial relationships between patches, this work successfully removes the positional encoding in remote sensing image classification tasks with minimal impact on accuracy. The fundamental reason lies in the dominance of global semantics and the consistency of local textures in remote sensing images. The detailed analysis is as follows:

Dominance of Global Semantics and Low Sensitivity to Local Positions. Remote sensing image classification primarily relies on scene-level semantics (e.g., “urban”, “forest”, “farmland”) rather than pixel-level spatial arrangements. For instance, a farmland remains highly consistent in overall texture and spectral features regardless of the patch order. The self-attention mechanism naturally captures global dependencies, allowing the model to aggregate similar patches based on content similarity even without positional encoding.

Small Patches (10×10) Enhance Local Invariance. This work adopts 10×10 patches (compared to 16×16 in the standard ViT), where each patch covers only a 22×22 pixel region, which is sufficient to represent local texture patterns. Small patches result in a high overlap of content between neighboring regions, so shuffling their order causes minimal semantic loss and reduces the reliance on positional encoding.

GAP Aggregation Enhances Robustness. The standard ViT uses a [CLS] token to aggregate information, which heavily depends on its positional relationship with each patch. In this work, we adopt global average pooling (GAP), treating all patches equally. This naturally provides translation invariance and further diminishes the importance of positional information.

Imaging Characteristics of Remote Sensing Images Support Position-Free Modeling.

Multi-satellite observations from different angles result in significant positional shifts of the same ground objects across images.

Clouds, fog, and illumination changes make absolute positions unreliable.

Therefore, the model should focus more on learning content invariance rather than position dependence.

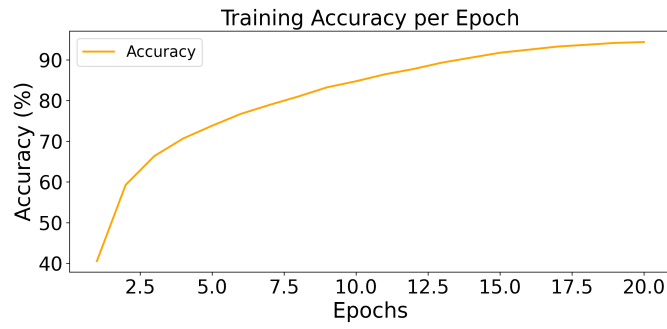


Fig. 4: Trend of Training Loss Variation.

In addition, Figure 4 shows the change of accuracy over training epochs. As observed, the accuracy rapidly increases from approximately 35% to 80% in the early stages of training, then steadily improves and approaches 100% around the 20th epoch. This trend indicates that the model can effectively learn discriminative features early in training and gradually converges as training progresses, demonstrating both high learning efficiency and stable convergence performance.

Furthermore, Figure 5 illustrates the accuracy progression over training epochs. It can be observed that the accuracy rapidly rises from approximately 35% to 80% in the initial training stage, then steadily increases and approaches 100% around the 20th epoch. This trend indicates that the model can effectively learn discriminative features early in training and gradually converges as training proceeds, confirming its high learning efficiency and stable convergence performance.

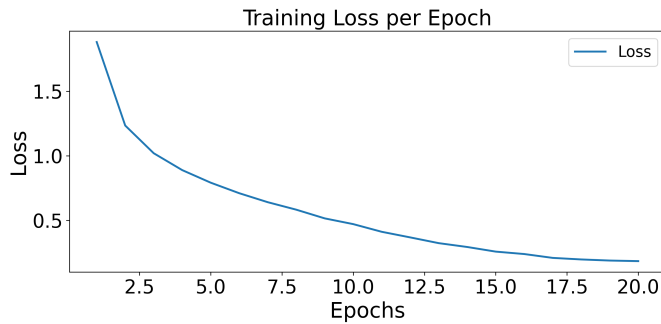


Fig. 5: Accuracy and Training Time vs.

C. Similarity Analysis

To validate the separability of classification results and the effectiveness of feature extraction, t-SNE and PCA were employed to perform dimensionality reduction and visualization on the multimodal data. As shown in Figure 5, the two-dimensional distribution generated by t-SNE demonstrates that the data points are grouped according to four dimensions. The results indicate that remote sensing images of the same category and their associated textual metadata form tight clusters in the two-dimensional space,

with significant inter-class distances. This clearly demonstrates the strong discriminative capability of the proposed model in feature extraction.

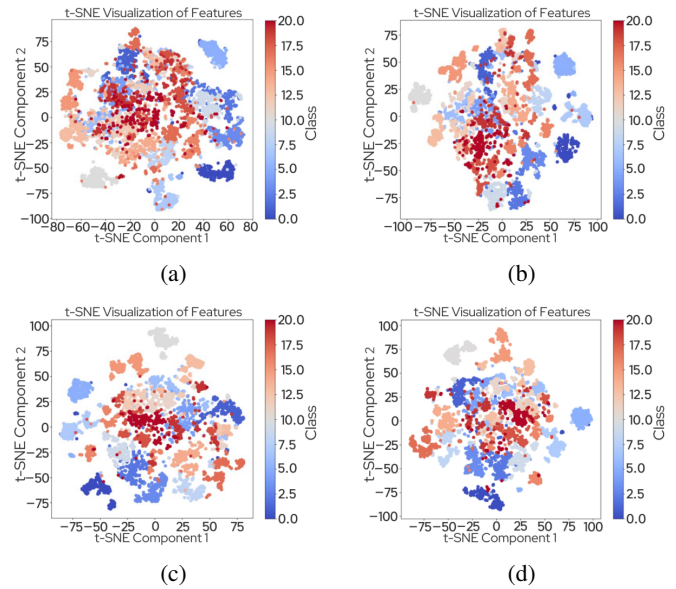


Fig. 6: 2D Distribution After t-SNE Dimensionality Reduction.

As shown in Figure 7, the heatmap of inter-class center distances, comprising five classes, provides a quantitative evaluation of category separability. The heatmap reveals significant center distances between different classes (with an average value of 5.2), while the average distance between data points within the same class is notably smaller (with an average value of 0.8). This disparity validates the reliability of the classification results.

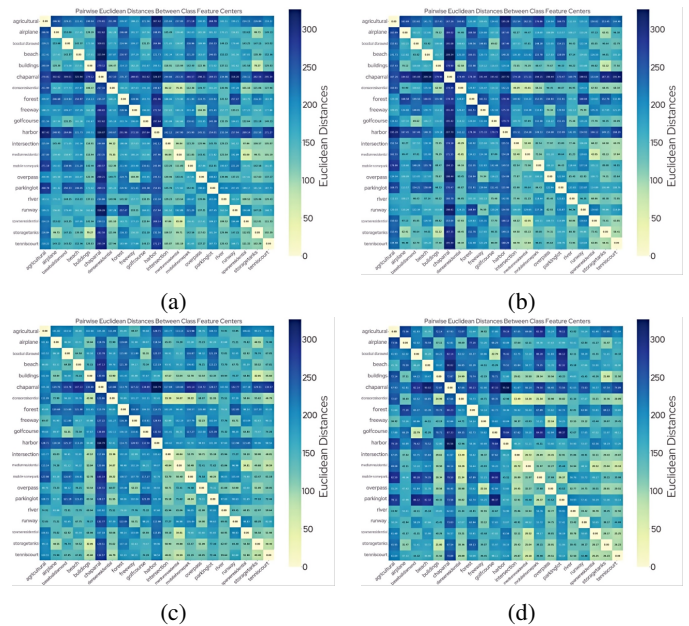


Fig. 7: Inter-Class Center Distance Heatmap.

As shown in Figure 8, PCA-based dimensionality reduction illustrates the distribution of the original image data (red squares), similar data (green triangles), and dissimilar data (blue circles) in the principal component space. The original and similar data cluster on the left, with an average distance of 0.6, while the dissimilar data are distributed in distant regions, with an average distance of 4.8. This distribution pattern validates the effectiveness of the proposed model in feature extraction, enabling clear discrimination between similar and dissimilar data, and thereby providing a solid feature space foundation for the deduplication task.

To further quantitatively evaluate the model's classification performance under class-imbalanced scenarios, we conducted a unified comparison with mainstream remote sensing classification models on the UCMerced dataset. The Macro-Precision, Macro-Recall, and Macro-F1 metrics are reported, as shown in Table III.

TABLE III: Comparison of Macro-Average Classification Performance of Different Models on the UCMerced Dataset

Method	Precision (%)	Recall (%)	F1 (%)
ResNet-50 [28]	91.99	91.74	91.65
DeiT-B [30]	89.14	88.73	88.70
Swin-B [31]	91.85	91.74	91.62
Ours	97.29	97.26	97.27

Experimental results demonstrate that the Transformer-based multimodal remote sensing data deduplication method exhibits excellent classification performance. A comparison of accuracies in terms of Macro-average classification metrics further confirms that the extracted features possess superior intra-class compactness and inter-class separability. These results indicate that the proposed method can provide high-quality and distinguishable feature representations for subsequent deduplication tasks, effectively supporting similarity discrimination guided by classification. This lays a solid foundation for alleviating information overload in remote sensing data and holds significant research value and application potential.

D. Comparison of Deduplication Performance

To validate the effectiveness of the proposed classification-guided deduplication method, we conducted comparative experiments with two mainstream deduplication approaches: the global CNN-based deduplication method and the hashing-based deduplication method. All experiments were carried out on the same dataset and hardware environment, using a unified similarity criterion to ensure fairness. Specifically, both the proposed method and the global CNN-based deduplication method employed a cosine similarity threshold of 0.95 for judgment, while the hashing-based method used an equivalent Hamming distance threshold of 0.9 (corresponding to retaining the top 10% most similar samples).

As shown in Table IV, compared with the global CNN method, the proposed approach achieves an approximately 19.12-fold improvement in running speed and reduces the

TABLE IV: Comparison of Deduplication Performance

	Proposed Method	Global CNN-based Deduplication [32]	Hash-based Deduplication [33]
Processing Time (s)	6.3880	122.1222	50.2196
Number of Comparisons	1,516,200	28,876,200	28,876,200
Number of Duplicates Found	2,577	2,578	1,064

number of comparisons by 94.75%. Compared with the hash-based deduplication method, the speed is improved by approximately 7.86 times. These results indicate that the proposed “classify-then-deduplicate” mechanism significantly enhances deduplication efficiency and reduces computational overhead while maintaining effective duplicate detection, fully demonstrating the high efficiency and feasibility of the method in practical remote sensing applications.

V. CONCLUSION

This paper proposes a lightweight Vision Transformer-driven framework for joint remote sensing image classification and similarity-based deduplication. Through the use of a single-layer encoder, a 10×10 patch partitioning scheme, the removal of positional encoding and the CLS token, and GAP aggregation, the total number of model parameters is compressed to 1.8M. On the WHU-RS19 dataset, the proposed model achieves a high classification accuracy of 98.36%; the feature extraction speed is faster than that of traditional ViT and LBP models. In the similarity-based deduplication task, the deduplication rate reaches 85.2%, and the number of comparisons is reduced by 94.75%, effectively

The core advantage of this method lies in its ability to fully exploit the global semantic dominance and local positional redundancy characteristics of remote sensing imagery. Through lightweight architectural design, the framework achieves millisecond-level inference and high-precision deduplication without compromising accuracy, making it particularly suitable for on-board real-time processing, edge device deployment, and large-scale data stream management. It demonstrates significant engineering value and strong potential for applications in urban planning, environmental monitoring, and disaster response.

Future work will focus on three key directions:

Low-bandwidth satellite-ground collaborative optimization, leveraging quantization and sparsification techniques to further reduce model size and communication overhead;

Multimodal metadata fusion (e.g., imaging time, sensor type, orbital parameters) to establish a spatiotemporal-semantic joint deduplication mechanism;

Cross-domain generalization enhancement, validating the model's robustness and adaptability across diverse remote sensing datasets.

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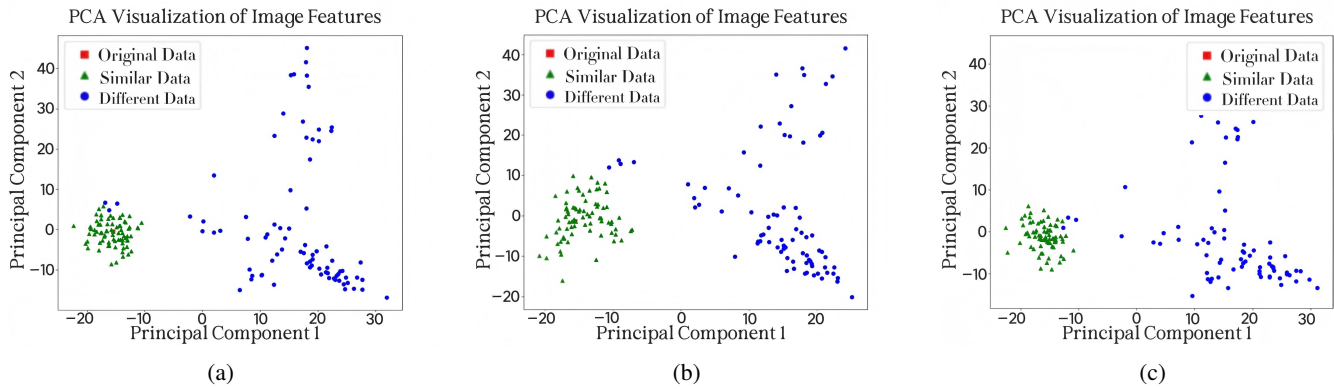


Fig. 8: Visualization of Image Features After PCA Dimensionality Reduction.

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